

IEEE 13-Node Feeder: Recloser–Fuse Coordination

Progress summary · DigSILENT PowerFactory 2021 SP2 · 27 September 2026

Reproducing M. Yousaf et al., “An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional Recloser and Fuse Coordination in Unbalanced Distribution Networks With Distributed Generation,” *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1831–1842, 2022.

At a glance

Item	Status	Result
Feeder model with 4.05 MVA DG at 692	Done	IEEE 13-node feeder in PowerFactory
Table II (branch currents)	Done	I _{nom} 100 %, I _{f,min} 92 %, I _{f,max} 83 % agreement with paper
Table III (fuse coefficients b _f)	Done	11 of 15 within 0.1 of paper; F633 off (6.43 vs 6.83)
Recloser R1 (GE IAC77B801A)	Done	Matches paper example to 0.3 %
Recloser R2	Stand-in	IAC used; paper uses GE Alstom CDG34
Time-overcurrent plot (R1, R2 fast + delayed)	Done	Curves match IAC formula to 0.05 %
PowerFactory crash on inserting plot	Worked around	Known dialog bug; project not corrupted
Fuses in the model	Not started	Table III computed from equations only

1. Feeder model and Table II

- IEEE 13-node test feeder built in PowerFactory with the synchronous DG at node 692 (Table I data).
- **I_{nom}**: unbalanced load flow, largest phase current at the protective-device end of each branch.
- **I_{f,max}**: bolted fault at the downstream node (3-phase, L-L or L-G by node phasing). **I_{f,min}**: L-G fault through 3 Ω at the farthest node. Complete method, DG out of service.

From	To	I _{nom} (A)	I _{f,min} (kA)	I _{f,max} (kA)
RG60	632	587.6	1.08	7.30
632	633	81.1	0.77	5.78
632	645	143.4	0.84	4.31
632	671	479.0	0.91	4.43
XFM1-HV	side	81.1	0.73	5.78
XFM1-LV	side	706.3	0.70	17.71
645	646	65.4	0.77	3.68
671	692	229.8	0.73	4.43
671	684	71.1	0.66	3.19
671	680	0.0	0.64	3.65
692	675	205.9	0.73	4.01
684	611	71.1	0.68	2.06
684	652	63.0	0.66	2.03

Average agreement with the paper: I_{nom} 100 %, I_{f,min} 92 %, I_{f,max} 83 %.

2. Table III: fuse coefficients

- Recloser pickup $I_p = 1.25 \times I_{nom}$ of the recloser branch; GE IAC extremely inverse curve $t = TDS \cdot (A + B/(M-C) + D/(M-C)^2 + E/(M-C)^3)$.
- $b_i = \log_{10}(t_F + i/(z+1) \cdot (t_D - t_F)) + 1.8 \cdot \log_{10}(I_i)$, with $a_i = -1.8$ (paper eq. 7 and 9).

Fuse	Recl.	i	z	I fuse (A)	I recl. (A)	t_F (s)	t_D (s)	b_i	b_i paper
F632	R1	1	2	4312	4479	0.091	1.815	6.37	6.38
F633	R1	1	2	5778	5876	0.062	1.239	6.43	6.83
F634	R1	2	2	17708	2311	0.276	5.517	8.22	8.32
F645	R1	2	2	4312	4479	0.091	1.815	6.64	6.65
F646	R1	2	2	3676	3888	0.112	2.247	6.60	6.67
F-DL	R1	1	1	4434	4464	0.091	1.824	6.55	6.54
F671	R2	1	1	3655	3720	0.141	0.706	6.04	6.08
F671-1	R2	1	3	4433	4434	0.110	0.550	5.91	5.85
F692	R2	2	2	4433	4434	0.110	0.550	6.17	6.13
F692-R	R2	2	3	4014	4037	0.126	0.628	6.06	6.14
F675	R2	3	3	4014	4037	0.126	0.628	6.19	6.19
F671-2	R2	1	3	3192	3365	0.164	0.820	5.82	5.87
F684	R2	2	3	2065	2242	0.319	1.597	5.95	6.15
F611	R2	3	3	2065	2242	0.319	1.597	6.07	6.19
F652	R2	2	2	2029	2246	0.318	1.592	6.02	6.17

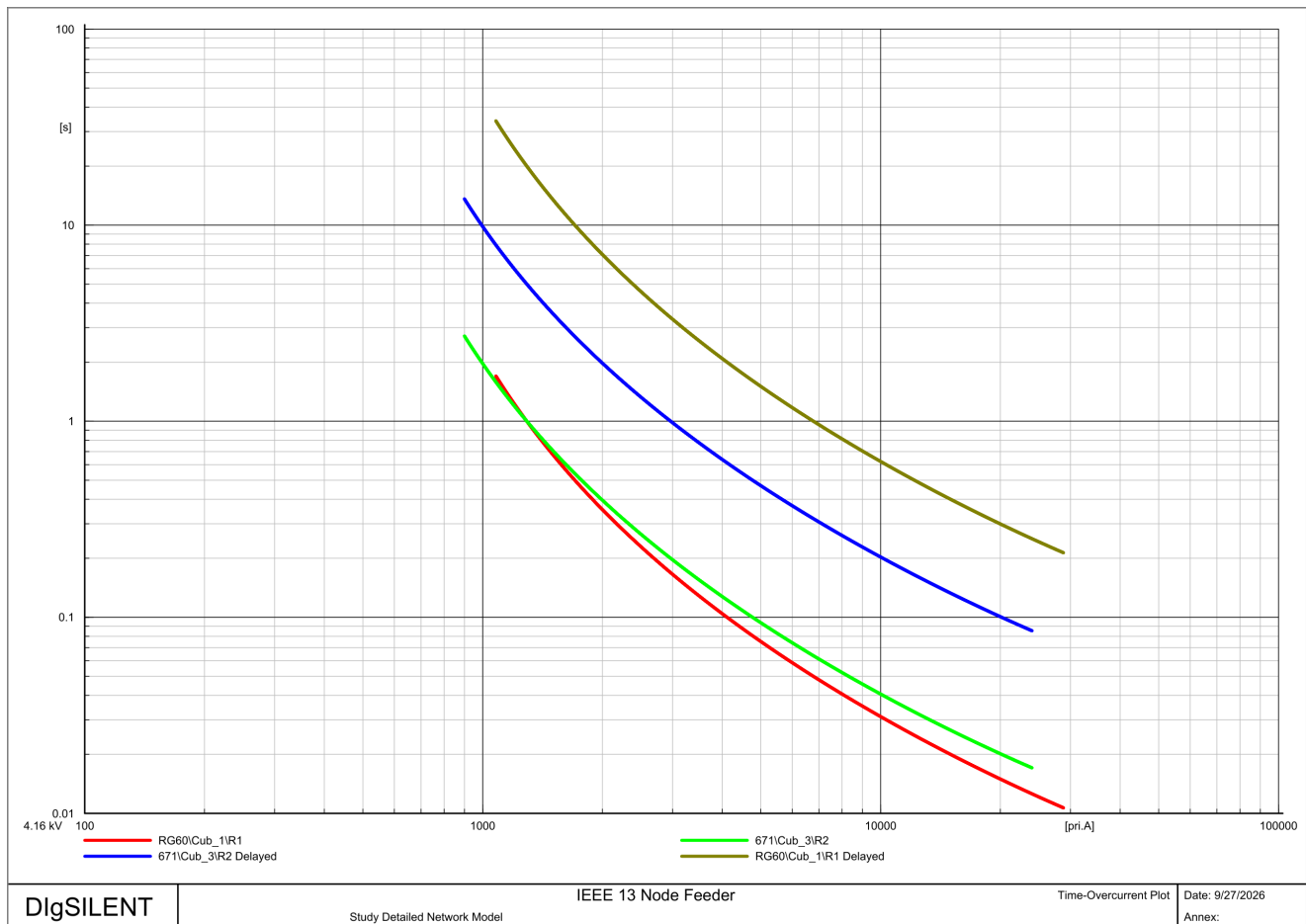
Highlighted rows differ from the paper by more than 0.1.

3. Reclosers R1 and R2

	R1 (feeder, RG60)	R2 (mid-line, 671)
Relay	GE IAC77B801A, extremely inverse	GE IAC77B801A (stand-in)
CT / tap	900/5, tap 4.0 A	1000/5, tap 3.0 A
Pickup	720 A (paper 734.5 A, -2.0 %)	600 A (paper-derived 598.8 A, +0.2 %)
Time dial fast / delayed	0.5 / 10 (from paper)	0.8 / 4.0 (fitted to Table III)
3-phase bolted fault	632: 7302 A $\rightarrow$ 0.045 s / 0.910 s	671: 4433 A $\rightarrow$ 0.110 s / 0.552 s

Check against the paper's worked example (4219 A below F645): R1 fast 0.096 s / delayed 1.926 s, against 0.097 s / 1.932 s in the paper.

4. Time-overcurrent plot



R1 fast (red), R2 fast (green), R2 Delayed (blue), R1 Delayed (olive). 4.16 kV, primary amperes.

Check	PowerFactory	IAC formula
R1 Delayed at 7302 A	≈ 0.9 s	0.910 s
R2 Delayed at 4433 A	≈ 0.55 s	0.552 s
R2 Delayed at 1968 A (cursor)	2.0375 s	2.0379 s
R2 Delayed at 1945 A (cursor)	2.0831 s	2.0840 s

- **Delayed curves coordinated:** R1 Delayed stays above R2 Delayed at all currents; margin about 1.2 s at the 4433 A fault at 671 (0.55 s vs 1.79 s).
- **Fast curves not coordinated:** R1 fast crosses below R2 fast at about 1300 A, so R1 trips 3–41 ms before R2 for faults below 671 between 1300 and 4433 A.

5. PowerFactory crash when inserting the plot

- All three crashes (27 Sep, 01:43–01:48) were the same access violation in PowerFactory's own dialog code (EditObject::StartEdit), triggered by the toolbar's insert-plot button.
- The project was not corrupted. PowerFactory would not start because a crashed PowerFactory.exe still locked the single-user database (error 4002). Ending it in Task Manager fixes this.
- Workaround: create plots by right-clicking a line or relay (Show → Time-Overcurrent Plot), or with Plot_Reclosers.py.

6. Open issues and next steps

Issue	Detail	Next step
R2 relay type	Paper uses GE Alstom CDG34; R2 fast too slow vs Fig. 8 (0.319 s bolted vs 0.207 s at 0.2 Ω)	Model CDG34 / IEC extremely inverse, $t = TMS \cdot 80 / (M^2 - 1)$; refit
R1–R2 fast coordination	R1 fast faster than R2 fast above ~1300 A	Refit R2 fast, or raise R1 fast time dial (~0.65)
Fuses	Not modelled; Table III from equations only	Add fuses fitting b_i ; plot each recloser with its fuses
F633 b_i	6.43 vs 6.83 in paper	Review fault current below F633
Fault on the plot	No fault marker yet	Run a short circuit with the plot open

7. Files

File	Purpose
IEEE 13 Node Feeder.pfd	PowerFactory project
IEEE13_Tables.py	Calculates Tables II and III; writes Table_II.csv, Table_III.csv
Configure_Reclosers.py	Sets up R1, R2 (fast + delayed relays, CTs, curves, time dials) and checks trip times
Check_Reclosers.py	Reports every relay's location, type, CT and stage settings
Plot_Reclosers.py	Builds the R1/R2 time-overcurrent plot without the crashing dialog